

The vector space model is an algebraic model for representing text documents as vectors of identifiers. It is used in information filtering, information retrieval, indexing and relevancy rankings. Documents and queries are represented as vectors in a high-dimensional space, where each dimension corresponds to a separate term in the vocabulary of the collection.

If a term occurs in the document, its value in the vector is non-zero. Several different ways of computing these values, also known as term weights, have been developed. One of the best known schemes is tf-idf weighting. The definition of a term depends on the application; typically terms are single words, keywords, or longer phrases.

Relevance rankings of documents in a keyword search can be calculated using the assumptions of document similarities theory, by comparing the deviation of angles between each document vector and the original query vector. Euclidean distance between the endpoints of two vectors is a poor choice, because it is large for vectors of very different lengths even when their content is semantically identical. Appending a document to itself, for example, produces a document with the same meaning but a large Euclidean distance from the original.

In practice it is easier to calculate the cosine of the angle between the vectors instead of the angle itself. A cosine value of zero means that the query and the document vector are orthogonal and have no match, that is, the query term does not exist in the document being considered. Ranking documents in decreasing order of the cosine of the angle is equivalent to ranking them in increasing order of the angle, since cosine is monotonically decreasing over the relevant interval.

The model has clear limitations. Long documents are represented poorly because they have low similarity values. Search keywords must match document terms precisely, so substrings and synonyms are missed. Documents with similar context but different vocabulary are not associated, which is the vocabulary mismatch problem that semantic retrieval later addressed. The order in which terms appear in the document is lost in the vector representation, since the model treats each document as a bag of words.